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What's special about 89% credibility intervals?

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A curious information-theoretic property of 89% credibility intervals is discussed.

A *credibility interval* (or "credible", "Bayesian", "posterior", "compatibility" interval or region) is a range where the true value of some real quantity lies with a given probability[1]. For instance, if $[1.2, 7.0]$ is a 60% credibility interval for the quantity Q, then there is a 60% probability that the true value of Q is between 1.2 and 7.0 (note the difference in meaning from a *confidence* interval, whose definition is much more involved and less direct[2]).

It is always best to report the full posterior probability distribution of a quantity of interest, so that the reader can calculate from it the probabilities of any desired ranges and other properties. But credibility intervals often provide useful summaries of the full distribution.

Credibility intervals of 50%, 75%, 90%, 95%, or even 99% probability are more or less commonly used in the Bayesian literature. Also *89%* ones have become quite fashionable since around 2016. They seemingly originate in McElreath's *Statistical Rethinking*[3], first published in that year. McElreath justly points out[4] that the specific probability value of a credibility interval should be chosen depending on the context; there is nothing special about values like 90% or 95%. To emphasize and remind us of this fact, he uses 89%, which is an arbitrary value.

Or is it? It turns out that 89% credibility intervals do possess a curious information-theoretic property.

The true value of a quantity lies inside a p credibility interval with probability p, and outside with probability $1 - p$. Our uncertainty about these two possibilities can be quantified by the Shannon entropy:[5]

$$H(p) \coloneqq -[p \log_2 p + (1-p) \log_2(1-p)]\,\mathrm{Sh}\,. \tag{1}$$

The value of 1 Sh (one shannon[6]) represents an equal uncertainty between two alternatives.

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1 Bernardo & Smith 2000 § 5.1.5 pp. 259ff. **2** MacKay 2005 § 37.3; Jaynes 1976. **3** McElreath 2020. **4** McElreath 2020 e.g.§ 3.2.2 p. 56, § 4.3.5 p. 88. **5** ISO 2008 Items 13-24, 13-25. **6** `https://electropedia.org/iev/iev.nsf/display?openform&ievref=171-07-11`

Thus for a 50% credibility interval we have an uncertainty of 1 Sh on whether the true value lies inside or outside it. This is the highest level of uncertainty for such dichotomous alternative. It is useful to also provide an interval with an uncertainty smaller than 1 Sh, so that we are somewhat more certain that the true value lies inside than outside.

For example, for which interval is our maximal uncertainty *halved* to 0.5 Sh? The answer is: for *an* 89.00% *credibility interval*. This result is easily calculated solving $H(p) = 0.5\,\mathrm{Sh}$ numerically. A more precise solution is 88.9972%. Vice versa the entropy corresponding to the exact probability 89% is approximately 0.499 916 Sh.

When we report a 50% and an 89% credibility interval we are therefore reporting intervals having uncertainties of one shannon and of half a shannon.

Of course the arbitrariness remains. We could also report intervals with uncertainties 1 Sh, $\frac{2}{3}$ Sh, $\frac{1}{3}$ Sh; their credibilities would be 50%, 82.6%, 93.9%. Or with uncertainties 1 Sh, 0.5 Sh, 0.25 Sh, corresponding to 50%, 89.0%, 95.8%.

For your amusement, some corresponding values of entropies and probabilities are reported in table 1.

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entropy/Sh	probability/%
1.00	50.00
0.95	63.09
0.90	68.40
0.85	72.40
0.80	75.70
0.75	78.55
0.70	81.07
2/3	82.60
0.65	83.33
0.60	85.39
0.55	87.27
0.50	89.00
0.45	90.59
0.40	92.06
0.35	93.42
1/3	93.85
0.30	94.68
0.25	95.83
0.20	96.89
0.15	97.85
0.10	98.70
0.05	99.44
0.00	100.00

probability/%	entropy/Sh	probability/%	entropy/Sh
50	1.0000	75	0.8110
51	1.0000	76	0.7950
52	0.9990	77	0.7780
53	0.9970	78	0.7600
54	0.9950	79	0.7410
55	0.9930	80	0.7220
56	0.9900	81	0.7010
57	0.9860	82	0.6800
58	0.9810	83	0.6580
59	0.9770	84	0.6340
60	0.9710	85	0.6100
61	0.9650	86	0.5840
62	0.9580	87	0.5570
63	0.9510	88	0.5290
64	0.9430	89	0.5000
65	0.9340	90	0.4690
66	0.9250	91	0.4360
67	0.9150	92	0.4020
68	0.9040	93	0.3660
69	0.8930	94	0.3270
70	0.8810	95	0.2860
71	0.8690	96	0.2420
72	0.8550	97	0.1940
73	0.8410	98	0.1410
74	0.8270	99	0.0808

Table 1 Some corresponding values of probabilities and entropies